

Technology & Scaling Plan

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Hamara MIS Dashboard live hai — Google Sheets ke data se poori company ka kaam (Task List, Checklist, multiple offices) track hota hai. **Data hamesha Google Sheet me hi rahega** (safe & in our control). Ye document 3 options deta hai taaki jaise-jaise company aur sheets badhein (aaj 3 → kal 50+), system utni hi speed & reliability se chale.

1. Abhi kya ready hai (delivered)

- **Executive Dashboard** — KPIs, department performance, status breakdown, sabki ek nazar me progress.
- **Task List & Checklist** — seedha Google Sheets se live, har doer ka kaam.
- **Done / Revise** — dashboard se hi task complete ya reschedule; wapas Google Sheet me likhta hai.
- **Multi-Sheet** — Admin panel se nayi Google Sheets easily connect (alag office / kaam ki alag list).
- **Access control** — konsi sheet kis doer ko dikhe, admin decide karta hai.
- **Cross-device** — assignment ek laptop pe karo, har device pe same dikhta hai.
- **Clean view** — sirf pending kaam dikhta hai; complete hote hi hat jaata hai (record Sheet me rehta hai).
- **Speed** — caching (server + browser) se repeat loads fast.
- **Login & roles** — Admin / PC / Employee accounts.

2. Core principle — Data Google Sheet me hi

Poora system is tarah banaya gaya hai ki **asli data Google Sheets me rahe** — kyunki wahan data safe hai, backup hota hai, aur team usse seedhe bhi kholker dekh/edit kar sakti hai. Neeche ke saare options is soch ka samman karte hain. Farak sirf itna hai ki data ko *kitni tezi se* aur *kitne bade scale* par serve kiya jaaye.

3. Aage ki soch (scaling ka sawaal)

Aaj 2–3 sheets hain — sab theek. Jab **50+ sheets** aur zyada log ek saath use karenge, tab "sheet ko baar-baar padhne" wala current tareeqa dheema pad sakta hai. Iske 3 raaste hain:

Option A

ABHI WALA

Kaise:

Google Sheet + Apps Script (+ caching, jo add kar diya gaya hai).

Speed:

Theek (cache se), par bhaari load par dheema.

50 sheets:

Apps Script ki limits (execution time, daily quota) अड़चन ban sakti hain.

Data:

Google Sheet me.

Kharcha:

Free.

Mehnat:

Ban chuka.

Faisla:

Aaj ke chhote scale ke liye badhiya. Badi company ke liye long-term nahi.

Option B

RECOMMENDED

Kaise:

Google Sheet (data wahi) + official **Google Sheets API** + Vercel serverless + ek cache layer . Apps Script hata dete hain.

Speed:

Kaafi fast — koi cold start nahi, koi 6-min limit nahi.

50 sheets:

Cache layer load & API-limits dono sambhaal leta hai.

Data:

Google Sheet me hi (100% same).

Auth:

Proper login.

Kharcha:

Free/sasta.

Mehnat:

Medium (one-time setup).

Faisla:

Data Sheets me rakhte hue sabse behtar balance — speed, scale & kam kharcha. Growth ke liye yahi.

Option C

ENTERPRISE

Kaise:

Asli database (Supabase / Postgres) + Google Sheet ko **auto-backup/export**.

Speed:

Instant, lakhon rows par bhi. **50/500 sheets:** koi farak nahi.

Data:

Database me (Sheet me auto-copy backup ke liye).

Auth:

Best (row-level security).

Kharcha:

Free tier bada, sasta.

Mehnat:

Zyada (migration).

Faisla:

Jab company bahut badi ho jaaye. Sabse fast & scalable, par tab data ka "asli ghar" database ban jaata hai (Sheet backup रहता hai).

4. Ek nazar me tulna

Kasauti	A — Abhi (Apps Script)	B — Sheets API + Cache	C — Database (Supabase)
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Speed	Theek	Fast	Sabse fast
50+ sheets	Atkega	Aaram se	Aaram se
Data Google Sheet me	Haan	Haan	Backup/export
Security / Login	Basic	Proper	Best
Kharcha	Free	Free / Sasta	Free tier / Sasta
Banane ki mehnat	Ho chuka	Medium	Zyada
Company scale ke liye	Nahi	Haan	Perfect

5. Recommendation (phased plan)

Abhi (0–3 mahine): Option A — jo ban chuka hai. Current scale ke liye kaafi. Kharcha zero.

Growth phase (jab sheets/users badhein): Option B par shift — *data Google Sheet me hi rakhte hue* speed & scale mil jaayega. Yehi hamari **sifarish** hai.

Enterprise (bahut badi company): Option C — database, Sheet auto-backup ke saath. Sirf tab jab data/users bahut zyada ho jaayein.

Ek line me: Data hamesha Google Sheet me safe rahega. Jab scale badhe, Option B se system 50+ sheets par bhi tez & reliable chalega — kam kharche me.

6. Kharcha (summary)

Teeno options **free ya bahut kam kharche** me chalte hain:

Sheets API — free

Supabase — free tier

Vercel — hosting (free/low)

Google Sheets — free

Koi bhi bada server ya license kharcha nahi. Scale badhne par hi mamooli cost aati hai.

7. Decision needed (Sir se)

- Kya hum **Option B** (Sheets API + Cache) par growth ke liye aage badhein? (*Data Sheet me hi rahega.*)
- Ya abhi **Option A** par hi rahein aur baad me review karein?

Approval milte hi ye step-by-step, bina kaam roke, implement kiya ja sakta hai.